

YAN-CHENG (BILL) HSU

Infrastructure Software Engineer | AI Platform & Kubernetes Expert

Sunnyvale, CA | bill.ych.jobs@gmail.com | linkedin.com/in/yan-cheng-hsu | bill-yc-hsu.com | Google Scholar (159+ citations)

PROFESSIONAL EXPERIENCE

Alibaba Cloud U.S. | Sunnyvale, CA

Infrastructure Software Engineer II | AnalyticDB Org, AI Training Platform | Jul 2025 – Present

Project 1 — Cross-Cluster AI Training Platform for Unitree humanoid robotics (Lead Engineer): multi-cluster control plane providing end-to-end heterogeneous compute, identity, and developer tooling for Unitree's training and simulation workflows.

- **Architected Heterogeneous Compute Platform:** abstracted heterogeneous compute (dedicated GPU + serverless CPU pools) across multiple Kubernetes clusters into a single resource substrate via **recursive K8s-on-K8s virtualization** (virtual-node-on-virtual-node); enabled Serverless-to-Reserved migration delivering **25x capacity scaling per availability zone** at **~40% TCO reduction** — the foundational compute layer all downstream SaaS on the platform depends on.
- **Engineered Cross-Cluster Identity Mesh:** extended the platform's K8s-on-K8s federation across **3 Kubernetes clusters** via application-layer routing + per-pod secrets-mount credential delivery — remote pods authenticate to customer VPC and private container registry transparently without static credentials.
- **Built AI Dev Workstations as a 0→1 SaaS Product on the Heterogeneous Compute Platform:** translated ML developers' contradicting needs — **secure access to customer's production data + remote IDE-grade iteration** — into a single-pod product abstraction. Pioneered dual-plane networking enabling simultaneous customer-VPC private egress AND public-internet developer ingress on the same pod — a capability the underlying cloud network model did not natively support; provisions in **<60s p95** vs days of DIY VPC peering + bastion setup. Led cross-stack delivery as Unitree's developer surface for interactive training and simulation.
- **Built Distributed Training & Simulation Scheduler as a 0→1 SaaS Product on the Heterogeneous Compute Platform:** abstracted ML teams' need to **mix CPU and GPU workers in a single training/simulation pipeline** into a clean user-facing job schema; per-component dispatch routes GPU-heavy workers to dedicated hardware and CPU-heavy workers to serverless infrastructure, **shielding users from underlying compute heterogeneity** while designed to deliver **~30-40% training cost reduction** on mixed-CPU/GPU workloads vs uniform GPU-only allocation. Led end-to-end product delivery.

Project 2 — RCAgent: Trust-First Multi-Agent Root-Cause-Analysis Platform (Lead Engineer, multi-agent kernel author): Alibaba-internal RCA system automating database on-call triage; currently delivering **~40% production alarm auto-healing** with verifiable diagnoses on confirmed-cause incidents.

- **Architected Supervisor-Worker Multi-Agent System with 4-Gate Hallucination Defense:** separated planning from execution via a supervisor agent dispatching to specialist sub-agents; new-skill additions gated by **pass³ ≥ 80%** (t-bench's pass^k consistency metric) on the regression suite — surfaces non-deterministic LLM failures missed by single-run tests, addressing a class of trust failures that pure-LLM agentic RCA systems systematically exhibit in production.
- **Designed Meta-Tool with Hierarchical Skill Tree:** agents perform multi-round retrieval over a 3-layer skill tree (deliberately simplified from a knowledge graph for tractability), narrowing the candidate set at each descent instead of registering every tool on each LLM call — yields **avg 6 tools loaded per invocation out of 200+ available** (~3% surface), **invariant to catalog growth**; lowers hallucination and bounds the agent's failure surface as new tools are added.

Amazon (AWS) | Seattle, WA

Software Development Engineer | High Performance Computing (HPC) & AI | Oct 2023 – Jul 2025

Core contributor to massive-scale GPU infrastructure: the backbone for training foundation models (Amazon Nova family).

- **Orchestrated GPU Lifecycle Management:** architected a scalable tracking system handling 3,000+ concurrent scaling events during peak training cycles; parent-child DAG in Apache Airflow eliminating circular termination loops, delivering ~\$9M annualized savings via optimized resource reclamation.
- **Automated Fault Remediation:** “divide-and-conquer” fault identification workflow for large-scale GPU clusters; dynamic K8s node labeling isolates and diagnoses hardware failures, reducing engineering troubleshooting time by ~90% per incident.

TECHNICAL SKILLS

- **Core:** Distributed Systems, System Design, Cloud-Native Architecture, High Availability, Multi-Agent Systems
- **Languages:** Golang (Expert), Python, C/C++, Java, TypeScript, SQL
- **AI Infrastructure:** Kubernetes (Internals, Operators, Virtual Kubelet, CNI/Pod Networks), Ray (Cluster/Serve), HPC, PyTorch Distributed, LLM Training Stacks
- **Cloud & Tools:** AWS (EKS, Fargate, Lambda, DynamoDB, VPC, Security Groups), Terraform, Prometheus, Grafana, eBPF

EDUCATION

M.S., Computer Science (Artificial Intelligence) | UC San Diego

2021 – 2023

B.S., Electrical Engineering | National Central University, Taiwan

2017 – 2020

PUBLICATIONS

- Hsu, Y.C. et al. “Generalized DNN Model for Cuffless Blood Pressure Estimation.” Sensors, 2020.
- Latifa, N.; Hsu, Y.C. et al. “On the Optimal Self-Supervised Multi-Fault Detector for Temperature Sensor Data.” APSIPA ASC, 2023.
- Amazon AGI Team. “The Amazon Nova Family of Models: Technical Report and Model Card.” 2024. (Contributor 4.1: **Bill Hsu**)